

Scalability & Future Plan

Date	09 JULY 2026
Team ID	XXXXXX
Project Name	EduGenie – AI Learning Assistant
Maximum marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The application currently supports a limited number of concurrent users.	Performance may decrease when many users access the system simultaneously.	High
2	SQLite database	Limited scalability	Medium
3	Internet dependency for Gemini AI	AI unavailable without internet	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users during testing.	Deploy the application on cloud platforms (Render, AWS, Azure, or Google Cloud) with load balancing to support thousands of users.
2	Data Storage	SQLite	Migrate to a relational database such as MySQL or PostgreSQL for efficient data management and scalability.
3	Performance	Standard processing	Optimize the model, use caching, and improve backend performance for faster response times.
4	Security	JWT Authentication	Add user authentication, HTTPS encryption, secure APIs, and role-based access control for enhanced security.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Voice-based AI Assistant.	Next 1 Month	Hands-free learning
Phase 2	Android Mobile App.	Next 3 Months	Better accessibility
Phase 3	Real-time Analytics Dashboard	Next 6 Months	Increase prediction accuracy and provide deeper analytical insights.
Phase 4	AI Career Mentor & Interview Preparation	Next 12 Months	Personalized career guidance